

**Digital Content Provenance:
A Focus Group Study on AI Labels
and Professional Identity Trust**

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TPINIA7006 Digital Content Provenance:

Path to Transparency & Authenticity in the Generative AI World

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I. Introduction

The market for AI-generated headshots has expanded rapidly in recent years, driven by the growing importance of digital identity and the availability of low-cost, accessible tools enabling users to generate professional images in minutes. While this accessibility reduces cost and effort, it also raises important questions about trust in digital representations.

In response to growing concerns about online trust, industry efforts have focused on standardized transparency signals. The Coalition for Content Provenance and Authenticity defines “Content Credentials” as “nutrition labels” that reveal an asset’s creation and modification history (C2PA, 2025). Supported by initiatives such as the Content Authenticity Initiative, these efforts assume that disclosure enables more informed user engagement. This study tests that assumption by examining how transparency around AI-generated headshots on hiring platforms shapes perceptions and engagement.

II. Literature Review

Transparency is widely believed to enhance trust in digital content, as providing information on how content is created and modified helps users interpret it more accurately and reduces deception (Paris and Donovan, 2019). Experimental studies also show that provenance signals can shape user judgments, sometimes increasing perceived transparency and trust in authenticity (Feng et al., 2023).

However, transparency signals do not always work as intended. Research shows that “AI-generated” labels can reduce perceived credibility but have limited impact on engagement (Wittenberg et al., 2025). Meanwhile, mislabeling can trigger similar penalties regardless of actual authenticity (Altay and Gilardi, 2024). The “implied truth effect” further suggests that

labeling some content may increase trust in unlabeled content (Pennycook et al., 2020). Overall, these findings indicate that audiences often respond more to labels than to the content itself.

Existing research has primarily focused on news and misinformation contexts, where users assess the truthfulness of content. In contrast, professional platforms like LinkedIn involve evaluating individuals rather than claims. As a result, it remains unclear whether transparency cues, such as “Generated by AI” labels, shape real-world decisions like messaging and hiring in the same way observed in prior studies.

II. Methodology

To address this literature gap, this study aims to probe whether AI labels act as decision-making heuristics, rather than transparency tools, in hiring contexts. The study focused on two research questions:

1. Does the “Generated by AI” label influence viewer engagement independent of whether the image is actually AI-generated or real?
2. Does the effect of the label vary across engagement contexts, from low-stakes messaging to higher-stakes hiring decisions?

The study employed a 2×2 factorial design to examine how image authenticity and label presence shape engagement decisions in hiring-related contexts. We conducted a 45-minute focus group with six participants in human resources, recruiting, or hiring-adjacent roles. To enhance representational diversity, the sample included individuals from varied racial backgrounds, age groups, and industries, comprising four male and two female participants.

During the focus group, participants were presented an array of real and AI-generated headshot images. The real images were collected from friends and family of our group, and the

AI-generated images were produced using Canva AI, with base images sourced from *This Person Does Not Exist*. The images were randomly distributed into four categories, with five images per category: (1) real images without a label, (2) real images labeled “Generated by AI” (misabeled-real), (3) AI-generated images without a label, and (4) AI-generated images labeled “Generated by AI.” This design isolates the effect of the label from the image itself. By comparing real images with and without labels, it identifies whether engagement is driven by actual authenticity or by the “Generated by AI” label alone.

Participants spent the first 20 minutes completing a Google Forms survey, indicating for each headshot whether they would reply on LinkedIn and consider the individual for an internship or junior role. The survey also collected participants’ general views on AI-generated headshots and labeling practices, following a 25-minute moderated discussion for deeper insights.

III. Findings & Analysis

Headshot Photo Survey Analysis

The two tables below present the survey results for two key questions: (1) whether participants would reply to the individual on LinkedIn, and (2) whether they would consider the individual for an internship or junior role.

	Real	AI-Generated	Row Difference
No Label	83.3%	60.0%	+23.3 pp
Labeled	73.3%	73.3%	0.0 pp
Column Difference	-10.0 pp	+13.3 pp	

Table 1: Reply on Question 1 - LinkedIn (percent “yes”)

	Real	AI-Generated	Row Difference
No Label	76.7%	53.3%	+23.3 pp
Labeled	66.7%	70.0%	-3.3 pp
Column Difference	-10.0 pp	+16.7 pp	

Table 2: Reply on Question 2 - Internship or Junior Role (percent “yes”)

According to the survey results, the participants interacted most frequently with non-labeled real headshots. This means focus group participants were more likely to reply on LinkedIn or consider the individual for a role when the individual had a real image with no label. Furthermore, when adding a label to a real image, the likelihood of interaction drops by 10pp in both instances. On the other hand, AI-generated headshots without a label received the least interaction. In the case of AI-generated headshots, adding a label increased the chance of replying on LinkedIn or consideration for a job opportunity by 13.3pp and 16.7pp respectively.

During the study, AI-generated headshots were initially presented without any labeling, and most participants responded positively, indicating that they would hire the individual and respond to them. As the slides progressed, each image was shown for 15 seconds, giving participants time to view the image and respond to the two evaluation questions. Once labels were introduced identifying the images as AI-generated, participant behavior noticeably shifted. Individuals began to examine the images more critically, and responses that were previously “yes” started to change to “no.” This suggests that awareness of AI involvement influenced perceptions of credibility and professionalism more than the images themselves.

These patterns suggest that labels play a significant role in shaping engagement decisions, often overriding the underlying authenticity of the image. While participants initially demonstrate a clear preference for real images, the introduction of labels reduces or eliminates this advantage, indicating that individuals rely more heavily on the presence of the label than on

the image itself. This effect is asymmetric: labeling real images introduces a penalty, reducing engagement even when the label is inaccurate, while labeling AI-generated images increases engagement, suggesting that disclosure may reduce uncertainty and make synthetic content more acceptable. Importantly, these findings do not suggest that transparency itself is ineffective, but rather that the *form* of transparency matters. The negative effects observed here appear to be driven by the limitations of binary “AI-generated” labels, which provide little context and may be interpreted as signals of risk rather than neutral information.

Opinions Survey Analysis

The other survey responses reveal a gap between low ability to detect AI-generated images and high concern about trust and authenticity. Participants overwhelmingly reported that they could not distinguish AI-generated from real headshots (100%) and expressed concerns about impersonation (83.3%). At the same time, attitudes toward AI headshots are mixed, with half viewing them as unprofessional and only a minority willing to hire candidates who use them (33.3%), while all respondents support labeling (100%).

These findings suggest that when users cannot reliably detect AI-generated images, they rely more heavily on external signals like labels to guide their decisions. However, mixed attitudes toward AI headshots indicate that transparency alone does not fully resolve concerns about trust and professionalism. This reliance on external signals highlights the importance of more informative transparency mechanisms. In environments where users cannot independently verify authenticity, simple labels may be insufficient, and more detailed provenance information, such as content credentials, may better support informed decision-making.

Focus Group Discussion Analysis

The survey findings are reinforced by the focus group discussion, where participants emphasized the difficulty of distinguishing AI-generated from real images, with one noting there was “no way...to tell whether that image was AI or not.” In this context, the “Generated by AI” label did not function as a neutral signal, but instead triggered suspicion and additional scrutiny. Participants described “apprehension” and questioned authenticity, interpreting the label as a signal of potential misrepresentation rather than simple disclosure. Importantly, this reaction reflects not a rejection of transparency itself, but dissatisfaction with how it is implemented, as several participants noted that a simple “AI-generated” label without additional context immediately prompted skepticism and was associated with deception or reduced effort. Together, these findings suggest that binary disclosure may unintentionally signal risk rather than honesty.

At the same time, participants expressed a more nuanced view of transparency. Rather than rejecting AI outright, they indicated that trust depends on how AI use is communicated, distinguishing between small enhancements and fully generated images. For example, one participant shared: “I’d want to know if AI was used heavily or just to touch up a background.” This suggests that transparency can support trust, but only when it conveys meaningful and interpretable information. Participants showed a clear preference for more detailed and contextualized disclosures that explain *how* AI was used, aligning with the concept of content credentials, which provide granular information about how an image was created and modified.

At the same time, participants emphasized that such disclosures must remain simple and easy to interpret. While there was interest in more detailed provenance information, users did not want overly complex explanations, instead preferring concise indicators that communicate key changes at a glance. One participant shared they wouldn’t have time for “the full story” but

would appreciate “some indicator that [the image] has been changed.” This highlights a critical design implication: effective transparency systems must balance informational depth with usability, ensuring that additional context enhances understanding without increasing cognitive burden.

IV. Conclusions: Takeaways & Recommendation

This study suggests that the “Generated by AI” label shapes engagement beyond simply conveying information about image provenance. Rather than reflecting authenticity, responses are guided by the presence of the label itself, which often overrides visual cues. Its impact is context-dependent, with minimal effects in low-stakes interactions and stronger influence in high-stakes decisions like hiring. In this sense, AI labels operate less as neutral transparency mechanisms and more as context-sensitive heuristics that structure how individuals make judgments.

Based on the findings, several design and policy recommendations emerge. For hiring platforms, we recommend they add content credentials for manual verification. However, since simple “AI generated” labels often trigger broad suspicion rather than informed judgment, platforms should move beyond binary labels and use more nuanced, interpretable categories such as “AI enhanced,” “AI assisted,” or “fully synthetic,” reducing overgeneralized distrust. Considering recruiters review many profiles quickly, simple visual cues, such as color-coded markers indicating different levels of AI use, can provide quick, intuitive information for fast verification. For job seekers, we recommend limiting AI use to minor enhancements (e.g., background adjustments) rather than full generation, as participants were more accepting of subtle edits.

To make these recommendations concrete, we built an interactive prototype that demonstrates how granular labels and content credentials could appear on LinkedIn profile cards. Readers can toggle between the current binary system and our proposed granular system to see how the leveling effect documented in our findings is mitigated under the proposed design. The prototype is available at <https://dprov-beyond-binary.netlify.app> and is described in Appendix A.

This study has several limitations that point to areas for future research. The sample is small and not fully representative, with participants drawn from similar academic and professional networks, limiting generalizability. Additionally, the focus group setting and simplified stimuli may not reflect real-world hiring decisions, where richer context such as resumes and networks shape evaluation. Finally, this study centers on evaluators, leaving open questions about how job seekers use AI-generated headshots and respond to recruiter engagement, which future research should explore.

This study hopes to contribute to a better understanding of how AI-mediated signals shape professional evaluation and to encourage further research on designing transparency mechanisms that support trust without distorting decision-making.

Word Count: 1978

References

- Aggarwal, R., Midha, V., & Sullivan, N. (2023). Effect of online professional network recommendations on the likelihood of an interview: A field study. *Information Systems Research*, 35(1), 104–119. <https://doi.org/10.1287/isre.2021.1053>
- Altay, S., & Gilardi, F. (2024). People are skeptical of headlines labeled as AI-generated, even if true or human-made, because they assume full AI automation. *PNAS Nexus*, 3(10), pgae403. <https://doi.org/10.1093/pnasnexus/pgae403>
- C2PA. (2025, May). *C2PA Technical Specification, Version 2.2*. Coalition for Content Provenance and Authenticity. https://spec.c2pa.org/specifications/specifications/2.2/specs/_attachments/C2PA_Specification.pdf
- Chesney, R., & Citron, D. K. (2019). Deep fakes: A looming challenge for privacy, democracy, and national security. *California Law Review*, 107(6), 1753–1819.
- Content Authenticity Initiative. (2025). *5,000 members: Building momentum for a more trustworthy digital world*. <https://contentauthenticity.org/blog/5000-members-building-momentum-for-a-more-trustworthy-digital-world>
- Gamage, D., Sewwandi, D., Zhang, M., & Bandara, A. K. (2025). Labeling synthetic content: User perceptions of label designs for AI-generated content on social media. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery. <https://doi.org/10.1145/3706598.3713171>

Härtel, T. M. (2025). (Don't) take me at face value: Decoding personality signals in LinkedIn profile pictures. *Personality and Individual Differences*, 246, 113297.

<https://doi.org/10.1016/j.paid.2025.113297>

Höltervennhoff, S., Ricker, J., Raphael, M. M., Schwedes, C., Weil, R., Fischer, A., Holz, T., Schönherr, L., & Fahl, S. (2026). “That's another doom I haven't thought about”: A user study on AI labels as a safeguard against image-based misinformation. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery. <https://doi.org/10.1145/3772318.3791006>

Jakesch, M., Hancock, J. T., & Naaman, M. (2023). Human heuristics for AI-generated language are flawed. *Proceedings of the National Academy of Sciences*, 120(11), e2208839120.

<https://doi.org/10.1073/pnas.2208839120>

Pennycook, G., Bear, A., Collins, E. T., & Rand, D. G. (2020). The implied truth effect: Attaching warnings to a subset of fake news headlines increases perceived accuracy of headlines without warnings. *Management Science*, 66(11), 4944–4957.

<https://doi.org/10.1287/mnsc.2019.3478>

Roulin, N., & Levashina, J. (2019). LinkedIn as a new selection method: Psychometric properties and assessment approach. *Personnel Psychology*, 72(2), 187–211.

<https://doi.org/10.1111/peps.12296>

Schiff, K. J., Schiff, D. S., & Bueno, N. S. (2024). The liar's dividend: Can politicians claim misinformation to evade accountability? *American Political Science Review*. Advance online publication. <https://doi.org/10.1017/S0003055423001247>

- Shi, L., & Viswanathan, S. (2023). Optional verification and signaling in online matching markets: Evidence from a randomized field experiment. *Information Systems Research*, 34(4), 1603–1621. <https://doi.org/10.1287/isre.2022.1194>
- Tifferet, S., & Vilnai-Yavetz, I. (2018). Self-presentation in LinkedIn portraits: Common features, gender, and occupational differences. *Computers in Human Behavior*, 80, 33–48. <https://doi.org/10.1016/j.chb.2017.10.013>
- Wittenberg, C., Péloquin-Skulski, G., Berinsky, A. J., & Rand, D. G. (2025). Labeling AI-generated media online. *PNAS Nexus*, 4(6), pgaf170. <https://doi.org/10.1093/pnasnexus/pgaf170>
- World Privacy Forum. (2024). *Privacy, identity and trust in C2PA: A technical review and analysis of the C2PA digital media provenance framework*. <https://worldprivacyforum.org/posts/privacy-identity-and-trust-in-c2pa/>

AI Disclosure

AI tools (Claude) were used to support literature search, refine English prose, and assist with document formatting. ChatGPT Deep Research was used to surface candidate citations, which were then reviewed and selectively integrated into the paper. The research design, focus group execution and moderation, data analysis, theoretical framing, and final arguments reflect the team's own work.

Appendix A. Interactive Prototype

We built an interactive prototype to demonstrate the granular disclosure system proposed in Section IV (Recommendations). The prototype is a static web page that allows readers to compare the current binary labeling system with our proposed four-tier system on a mock LinkedIn profile grid.

URL. <https://dprov-beyond-binary.netlify.app>

Design choices. Eight profile cards span four provenance tiers (Verified Human, AI Enhanced, AI Assisted, Fully Synthetic), with two cards per tier. Half of the cards carry the binary “Generated by AI” label in the current-system mode, including two cards whose underlying images are authentic. This mirrors the mislabeled-real condition from our focus group methodology and lets readers experience the leveling effect first-hand before switching to the proposed view. Each card opens a Content Credentials panel that mirrors the C2PA open standard provenance chain.

Image source. All eight portraits are AI-generated samples from thispersondoesnotexist.com, consistent with the base-image methodology used in our focus group stimuli. No real persons are depicted in the prototype.

Implementation. The prototype is a static page (HTML, CSS, vanilla JavaScript) deployed on Netlify. Source code: contact the lead author.